Quasi-Categories vs. Complete Segal Spaces

Joseph Rennie

2/9/2017

In this talk we compare two other models of higher categories: quasi-categories and complete Segal spaces. Actually, we will show that there are two different equivalences, that are inverses of each other.

Motivation

Last time we made our first effort to relate different models of higher categories to each other by comparing quasi-categories and simplicial categories ([Ba17]). This time we want to compare quasi-categories and complete Segal spaces. Before we do so we might wonder why we would even work with two different models? Both of them work with simplicial objects (sets or spaces) and thus seem quite similar. Still, it turns out every model has its own advantages and disadvantages and working with different models enables us to do various things. Here is a list of different benefits of both models to see that each are valuable on their own.

Quasi-Categories	Complete Segal Spaces
1) Can use combinatorial methods	1) Simplicial Model Structure
2) Easy to compare to simplicial categories	2) Can use localizations
3) Easier collection of generators	3) Generalizes to Complete Segal Objects
4) More work available	4) Can use homotopical methods

Thus we are certainly justified in working with different models based on our need.

Background

We discussed complete Segal spaces at great lengths in [Fa17]. It comes attached with a model structure, namely the complete Segal space model structure, which is the localization of the Reedy model structure. More generally we have following table of model structures and fibrant objects:

Name	Fibrant Objects:
Reedy Model Structure	Reedy Fibrant Objects
Segal space model Structure	Segal Spaces
Complete Segal Space (Rezk) Model Structure	Complete Segal Spaces

We move from the Reedy model structure to the Segal space model structure by inverting the spines using Bousfield localization:

$$G(n) = F(1) \coprod_{F(0)} ... \coprod_{F(0)} F(1) \to F(n)$$

we get from the Segal space model structure to the complete Segal space model structure by inverting the "walking isomorphism"

$$F(0) \to F(1) \coprod F(3) \qquad \coprod_{F(1) \coprod_{F(0)} F(1) \coprod_{F(0)} F(1)} F(1)$$

For more details on this see Section 7 of [Fa17].

We also discussed quasi-categories, which also has a model structure, namely the Joyal model structure:

Theorem 2.1. *There is a model structure on $s\mathcal{S}et$, with*

- *Fibrant objects: quasicategories;*
- *Weak equivalences: categorical equivalences.*

Details on the model structure and other details can be found in [Pa17].

Before we move on we will review one last theorem that will show up several times. Recall that two model structures are equivalent if there is a Quillen equivalence between them (or at least a zig zag). There are several ways to show a map is a Quillen equivalence. We will refer to following theorem for that (Section 2 of [Sh17])

Theorem 2.2. *A Quillen adjunction $F : \mathcal{M} \rightleftarrows \mathcal{N} : G$ is a Quillen equivalence if and only if the following two conditions are satisfied:*

1. *For every fibrant-cofibrant object $A \in \mathcal{M}$, the derived unit map $A \to GF(A) \to GR(F(A))$ is a weak equivalence in $\mathcal{M}$. Here $R(F(A))$ is the fibrant replacement of $F(A)$ inside $\mathcal{N}$ (it is a homotopy colocalization).*
2. *For every fibrant-cofibrant object $B \in \mathcal{N}$, the derived unit map $FQ(G(B)) \to FG(B) \to B$ is a weak equivalence in $\mathcal{N}$. Here $Q(G(B))$ is the cofibrant replacement of $G(B)$ inside $\mathcal{M}$ (it is a homotopy localization).*

We also have following lemma about Quillen equivalences that we will need in the last part of the proof.

Lemma 2.3. *(2 out of 3 for Quillen Equivalences) Let*

$$\mathcal{M} \underset{G}{\overset{F}{\rightleftarrows}} \mathcal{N} \underset{I}{\overset{H}{\rightleftarrows}} \mathcal{O}$$

be a chain of Quillen adjunction between model categories $\mathcal{M}, \mathcal{N}$ and $\mathcal{O}$. If from the set of Quillen adjunctions $\{(F,G),(H,I),(HF,IG)\}$ two are Quillen equivalences, then so is the third one.

The Quillen Equivalences

So, now that we have reviewed the relevant material and have a reasonable motivation we will prove the main theorems. Our goal is it so prove there are two Quillen equivalences between the Joyal model structure and the CSS model structure:

$$(s\mathcal{S}\mathrm{et})^{Joy} \underset{i_1^*}{\overset{p_1^*}{\rightleftarrows}} (s\mathcal{S})^{CSS} \underset{t^!}{\overset{t_!}{\rightleftarrows}} (s\mathcal{S}\mathrm{et})^{Joy}$$

We will first discuss the left hand adjunction and show it is a Quillen equivalence. Then we come back and show that the other side is also an equivalence using the first one. All the definitions and results here are originally from [JT07]. However, although all proofs are from that paper, we will rely on our own notation, which we introduced in [Sh17], [Pa17] and [Fa17].

The (p_1^*, i_1^*)-Adjunction

We first start with the adjunction

$$(s\mathcal{S}\mathrm{et})^{Joy} \underset{i_1^*}{\overset{p_1^*}{\rightleftarrows}} (s\mathcal{S})^{CSS}$$

First we have to actually define the functors (the following definition is from the beginning of Chapter 4 in [JT07]).

Definition 4.1. There is an adjunction

$$s\mathcal{S}\mathrm{et} \underset{i_1^*}{\overset{p_1^*}{\rightleftarrows}} s\mathcal{S}$$

The left adjoint p_1^* takes a simplicial set and embeds it horizontally. Thus $p_1^*(S)_{nl} = S_n$. The functor i_1^* takes a bisimplicial set X and gives us the first row $i_1^*(X)_n = X_{n0}$.

By using the given assumptions and doing lifting Calculus we can show that this pair is a Quillen adjunction. (first part of Proposition 4.7 in [JT07]).

We will use that and prove it is a Quillen equivalence by proving that it is a homotopy localization and colocalization. We will need following lemmas along the way:

Lemma 4.2. *Let X be a CSS. Then for every $k \in \mathbb{N}$, the row $X_{\bullet k}$ is a quasi-category (Corollary 3.6 of [JT07]). Moreover for two $k_1, k_2 \in \mathbb{N}$, any simplicial map $X_{\bullet k_1} \to X_{\bullet k_2}$ is a categorical equivalence (Proposition 3.9 of [JT07])*

Lemma 4.3. *Let $f : X \to Y$ be a map of simplicial spaces. If f is a row-wise categorical equivalence, then it is a CSS equivalence.*

Now we are in a position to prove that our adjunction is a localization. For that we have to show that for every CSS X, our derived counit map

$$p_1^* Q i_1^* X \to X$$

is a CSS equivalence (here Q is a cofibrant replacement). But every object is cofibrant and so we can just look at the map

$$p_1^* i_1^* X \to X$$

However, by definition $(p_1^* i_1^* X)_{kn} = (p_1^* i_1^* X)_{k0}$. But all the rows are categorically equivalent and so we have a row-wise categorical equivalence. This argument is the second part of Proposition 4.7 of [JT07].

Now for the homotopy colocalization. We have to show that for every quasi-category S, the map

$$S \to i_1^* R p_1^* S$$

is a categorical equivalence, where $R p_1^* C$ is a CSS fibrant replacement of $p_1^* S$.

For this we will actually go through the effort of finding an interesting fibrant replacement of $p_1^* C$, which does not modify the 0-row. We do that by generalizing the "classifying diagram" as introduced by Rezk in Section 3 of [Re01] (also reviewed in Section 3 of [Fa17]). For each quasi-category S we define the simplicial space $\Gamma(S)$ as

$$\Gamma(S)_m = (S^{\Delta[m]})^{core}$$

This generalizes the classifying diagram as follows. If $S = NC$ the nerve of a category, then $\Gamma(NC) = \mathcal{N}C$, so it generalizes the nerve construction from categories to quasi-categories. Similar to the case of categories, we have following lemma:

Lemma 4.4. *If S is a quasi-category then $\Gamma(S)$ is a complete Segal space.*

The proof can be found in Proposition 4.10 of [JT07] and is a big exercise in showing that the inner horn lifting conditions can be translated to the Reedy, the Segal and completeness condition. From the lemma above we also clearly see that the map

$$p_1^* S \to \Gamma(S)$$

is a CSS equivalence, as it is a row-wise categorical equivalence. Actually, all we have to check is the first row as $p_1^* S$ is trivial in the higher rows and at level 0 the map is just the identity.

Remark 4.5. It's worth reiterating the last point. $\Gamma(S)$ does not modify S in the first row. In other words, $\Gamma(S)_{n0} = S_n$.

Now we can prove that the adjunction above is a homotopy localization. For that we have to show that

$$S \to i_1^* R p_1^* S$$

is a categorical equivalence. But we just showed that $Rp_1^* C = \Gamma(S)$, so we have to show that

$$S \to i_1^* \Gamma(S)$$

is an equivalence. But by the remark above $i_1^*\Gamma(S) = S$, so the derived unit map is just the identity map! (this final step comes from Theorem 4.11 in [JT07]) Thus, we have proven that our adjunction

$$(s\mathcal{S}\mathrm{et})^{Joy} \underset{i_1^*}{\overset{p_1^*}{\rightleftarrows}} (s\mathcal{S})^{CSS}$$

is actually a Quillen equivalence.

The $(t_!, t^!)$-Adjunction

Now we move on to the second adjunction. We want to show that the following adjunction

$$(s\mathcal{S})^{CSS} \underset{t^!}{\overset{t_!}{\rightleftarrows}} (s\mathcal{S}\mathrm{et})^{Joy}$$

is a Quillen equivalence. But, first we have to define the adjunction. Before that we will need one last definition.

Definition 5.1. Let $I[n]$ be the category with n objects and one unique isomorphism between each two objects. Now, let $\Delta[n]' = N(I[n])$. Then $\Delta[n]'$ is a Kan fibrant replacement for $\Delta[n]$.

Definition 5.2. There is an adjunction

$$(s\mathcal{S}) \underset{t^!}{\overset{t_!}{\rightleftarrows}} (s\mathcal{S}\mathrm{et})$$

defined as follows. $t_!(F(n)) = \Delta[n]$ and $t_!(\Delta[n]) = \Delta[n]'$. Now we just define $t_!$ as the left Kan extension of the map, using the fact that $s\mathcal{S}$ is presentable and $F(n) \times \Delta[l]$ generates all of $s\mathcal{S}$

Remark 5.3. Based on this description, the right adjoint, $t^!$, is just defined as:

$$(t^!(S)_{nl} = Hom(F(n)\times\Delta[l], t^!(S)) = Hom(t_!(F(n)\times\Delta[l], S) = Hom(\Delta[n]\times\Delta[l]', S)$$

For more details on this definition see Page 13 of [JT07]. The fact that this adjunction is a Quillen adjunction follows from a combination of Theorem 2.12, Theorem 3.3 and the first part of Theorem 4.12 of [JT07]). All that remains is to show that it is a Quillen equivalence. here we will use 2 out of 3 for Quillen equivalences. In the chain

$$(s\mathcal{S}\mathrm{et})^{Joy} \underset{i_1^*}{\overset{p_1^*}{\rightleftarrows}} (s\mathcal{S})^{CSS} \underset{t^!}{\overset{t_!}{\rightleftarrows}} (s\mathcal{S}\mathrm{et})^{Joy}$$

we already showed the left hand side is a Quillen equivalence. Moreover for each simplicial set S, we have

$$t_! p_1^* S = t_! p_1^* \varinjlim_I \Delta[n] = \varinjlim_I t_!(p_1^* \Delta[n]) = \varinjlim_I (t_! F(n)) = \varinjlim_I \Delta[n] = S$$

Thus $t_! p_1^* = id$ and so $i_1^* t^!$ is isomorphic to the identity as well. Hence, it must be a Quillen equivalence. Using 2 out of 3, we get that $(t_!, t^!)$ is also a Quillen equivalence (This is the argument in Theorem 4.12 of [JT07]).

Remark 5.4. Last time we discussed how there is only one non-trivial automorphisms (which inverts the arrow) in any model of higher categories, as proven in [To05]. We can really see this fact in action in the last proof. We are composing two Quillen equivalences and we didn't really invert any direction with those maps and so we had to end up with the identity, which we did.

Upshot

The fundamental implication of these Quillen equivalences is that although complete Segal spaces have "2 axes", only one of them, namely the horizontal one. That is because the vertical axis has only homotopical information (essentially directionless information), which is completely subsumed by the horizontal axis, which has the categorical information.

A quasi-category is essentially the minimum amount of data you can take from a complete Segal space, without losing any higher categorical information. Even though no fundamental information is lost, some of the finer properties (such as being simplicial) is irretrievably gone.

References

[Ba17] W. Balderrama, *Quasi-Categories vs. Simplicial Categories*, Unpublished notes, http://math.illinois.edu/ rasekh2/seminars/qcatvsscat.pdf.

[Be10] J.E. Bergner, *A survey of (∞,1)-categories*, The IMA Volumes in Mathematics and its Applications. Springer, 2010. pp. 69-83

[Fa17] M. Fassina, *Introduction to CSS*, Unpublished notes, http://math.illinois.edu/ rasekh2/seminars/css.pdf.

[JT07] A. Joyal, M. Tierney, *Quasi-categories vs Segal spaces*, Contemp. Math 431 (2007): pp. 277-326

[Lu09] J. Lurie, *Higher Topos Theory*, Annals of Mathematics Studies 170, Princeton University Press, Princeton, NJ, 2009, xviii+925 pp. A

[Pa17] A. Panagiotopoulos, *Review of Quasi-Categories* Unpublished Notes http://math.illinois.edu/ rasekh2/seminars/quasicat.pdf

[Re01] C. Rezk, *A model for the homotopy theory of homotopy theory*, Trans. Amer. Math.Soc., 353(2001), no. 3, 973-1007.

[Sh17] B. Shin, *Review of Model Categories*, Unpublished notes, http://math.illinois.edu/ rasekh2/seminars/modelcat.pdf.

[To05] B. Toën, *Vers une axiomatisation de la théorie des catégories supérieures*, K-Theory, 34 (2005), no. 3, 233-263.